Lesson 5: Experimental Methodology

Technologies for Artificial Intelligence

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Before we start

- Exercise 4, which we discuss now
- The threshold you chose, and how you chose it

We broke a promise on purpose

- Lesson 4 chose a threshold **by looking at the test set**
- Lesson 1 forbade exactly that
- Today we pay the debt

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Today: how to tell whether it works

- One split is a lottery, and how wide the lottery is
- Cross-validation: what it estimates, and what it does not
- Bias, variance, and the floor you cannot go below
- Learning curves: which fix to buy
- Leakage that survives cross-validation

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A test score is a measurement

- You would not report a poll of forty people to three decimals
- Holding data out makes the estimate **unbiased**
- It does nothing to make it **precise**

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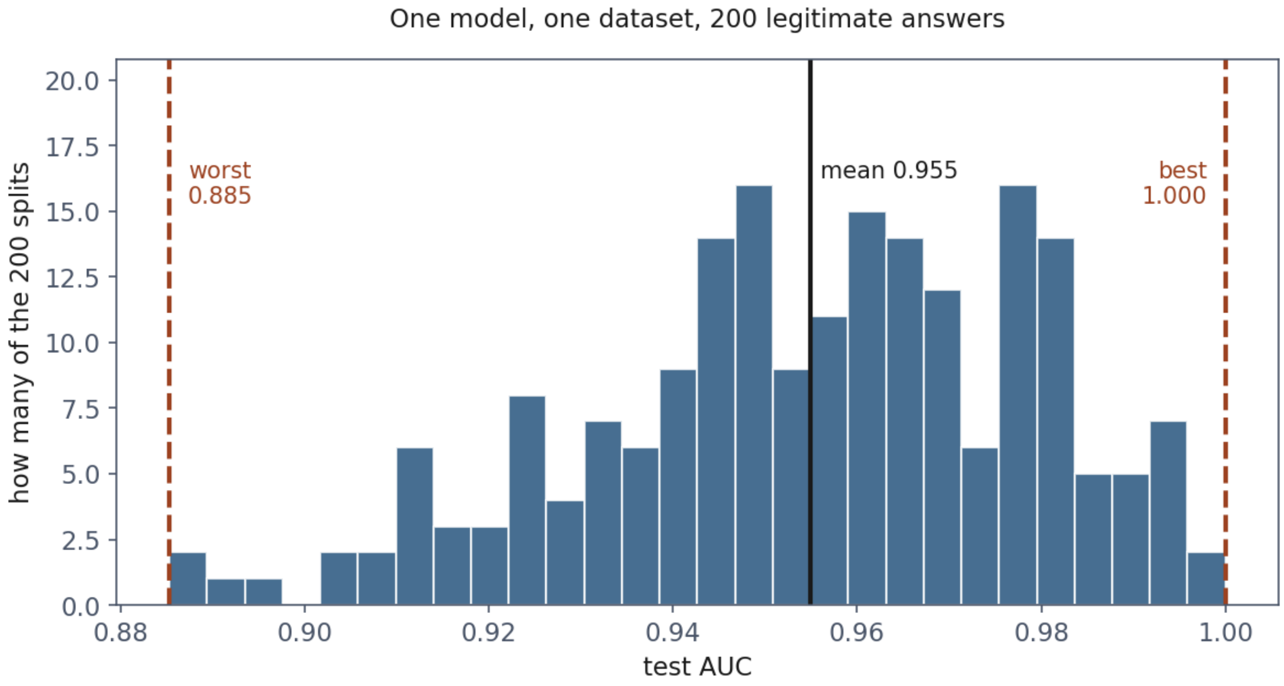
The fleet, cut down

- Lesson 4’s drives, reduced to **800**, of which **29 fail**
- Not to exaggerate: 800 rows with 29 positives is the ordinary case

200 honest splits, one dataset



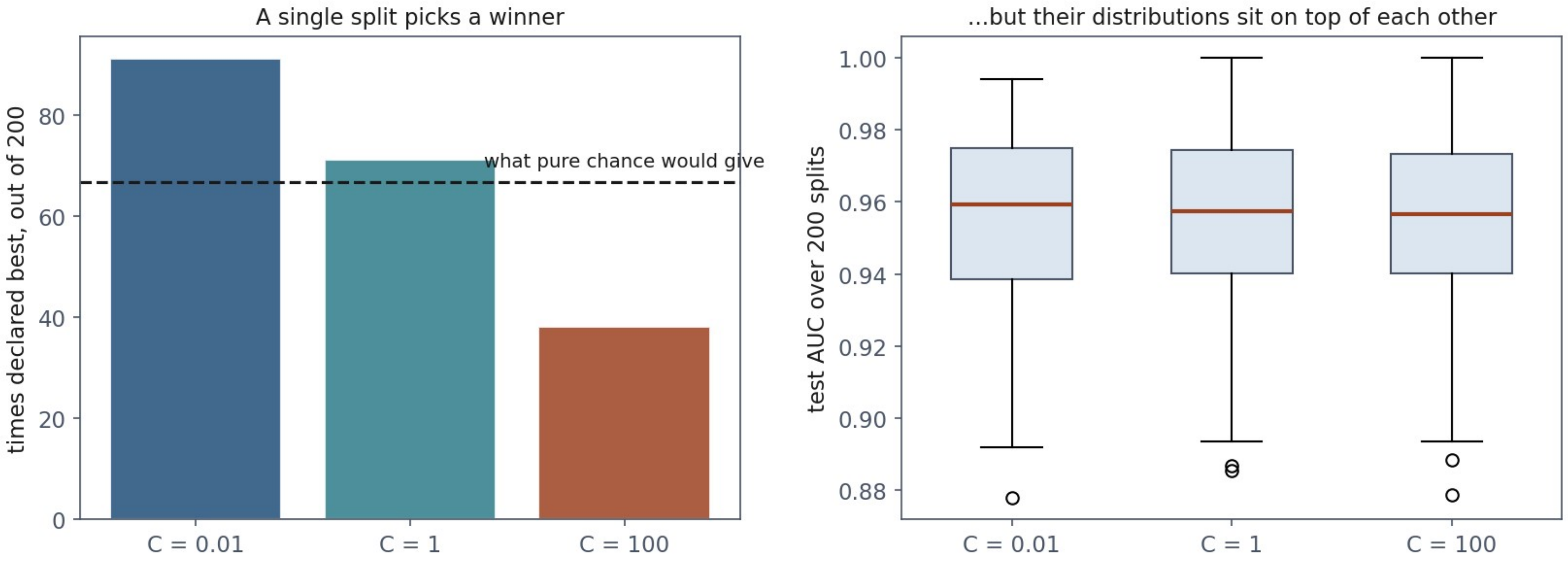
Why so wide?

- Each test set holds **7 failures**
- Every AUC (area under the receiver operating characteristic curve) rests on seven

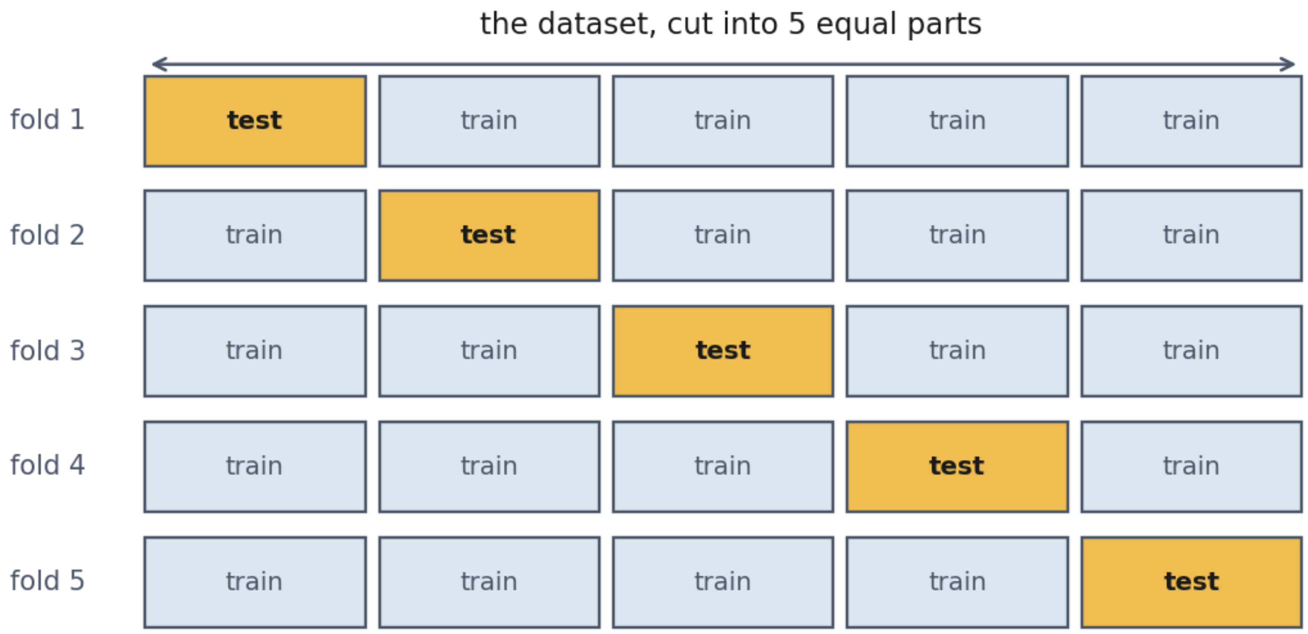
Now the expensive part: choosing

Model	Mean AUC	Times declared best
C = 0.01	0.9549	91
C = 1	0.9549	71
C = 100	0.9544	38

The same three models, drawn honestly



k-fold cross-validation



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What it actually estimates

- The performance of a **procedure**, not of one fitted model
- Slightly **pessimistic**: each fold trains on 80% of the data

Report the spread, but not as a confidence interval

- Any two training folds share **three quarters** of their rows
- The scores are correlated, so the usual standard error is optimistic

Stratify when a class is rare

Fold	Plain KFold	StratifiedKFold
1	7 failures of 160	5 of 160
2	**1 failure of 160**	6 of 160
3	7 of 160	6 of 160
4	5 of 160	6 of 160
5	9 of 160	6 of 160

How much does it help?

	Single split	5-fold CV
worst	0.9119	0.9439
best	1.0000	0.9588
spread	0.0881	0.0149

How many folds?

- **k = 5 or 10** in practice
- Larger k: less pessimistic, more expensive, folds more correlated
- k = m is **leave-one-out**

When the rows are not independent

- Ten readings from the **same drive**, split at random
- Nine in training, one in test
- The model recognises the drive, not the failure

Split by the thing you must generalise to

- Same entity in two folds → **GroupKFold**
- Predicting the future → **TimeSeriesSplit**

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Notebook 1, live

- The split lottery, the three identical models, k-fold by hand

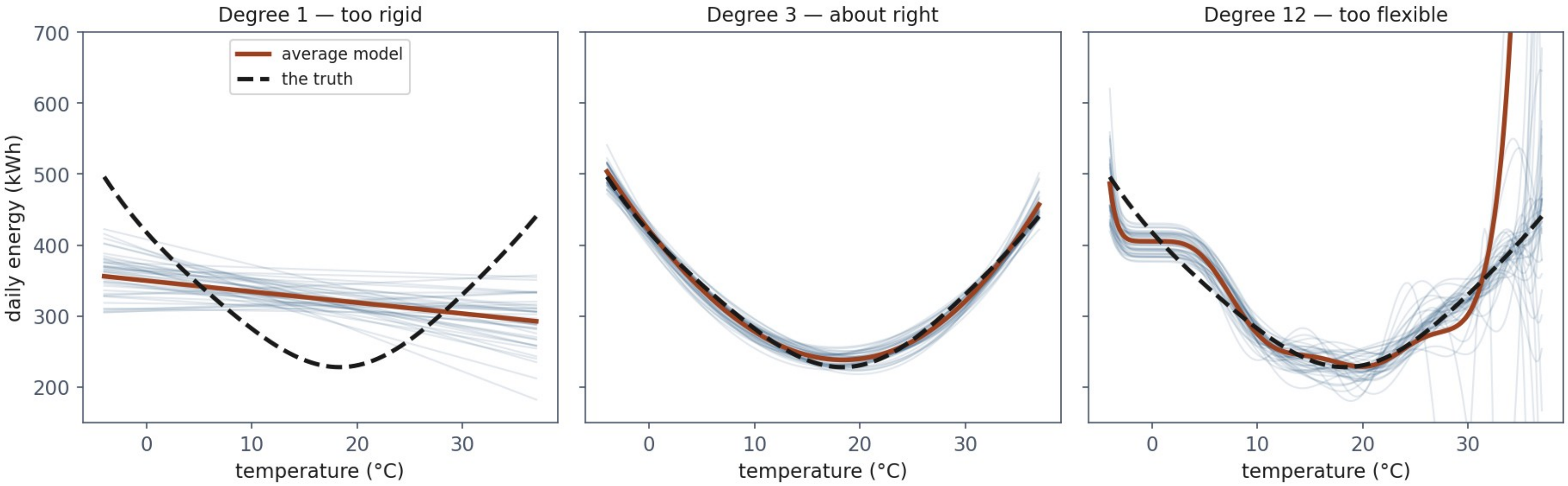
Break

- Ten minutes

Where does the error come from?

- **Bias**: the average model is wrong
- **Variance**: the individual models disagree
- **Noise**: the target itself is uncertain

Three hundred parallel universes



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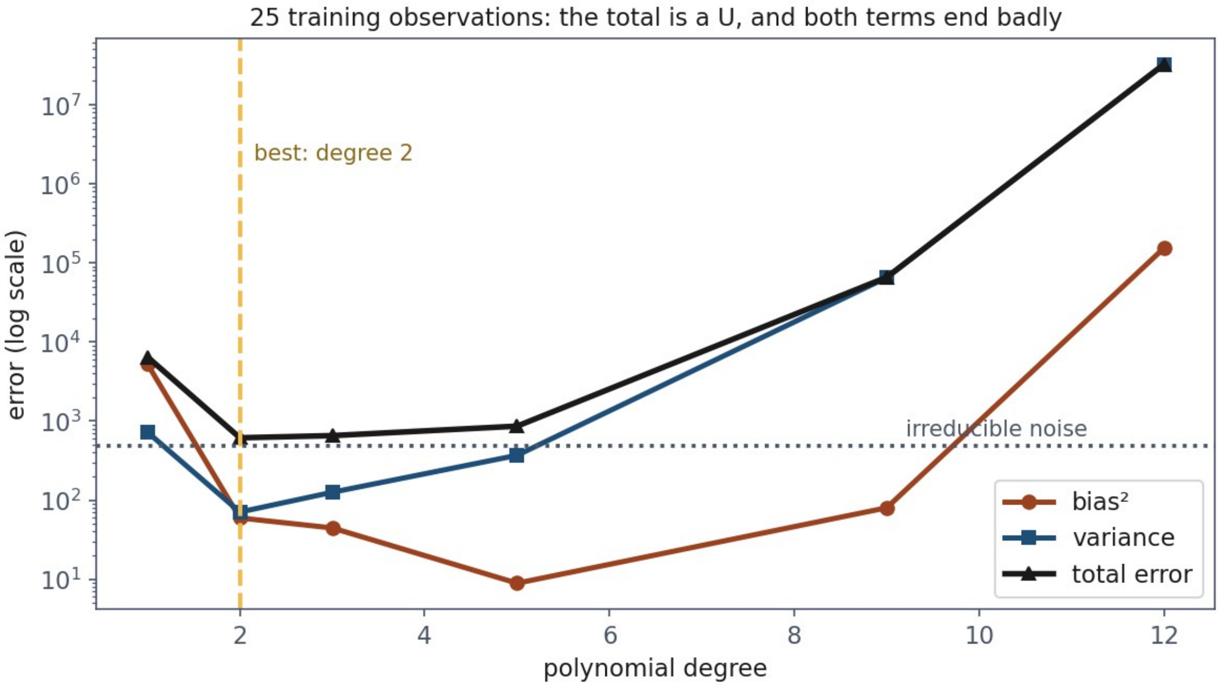
The decomposition

$$\mathrm{E}\left[(y-\hat{f})^2\right] = \mathrm{bias}^2 + \mathrm{variance} + \sigma^2$$

Measured, on 25 observations

Degree	$bias^2$	Variance	Noise	Total
1	5,250.8	723.8	484.0	6,458.6
2	59.6	70.8	484.0	**614.3**
5	8.9	368.9	484.0	861.8
12	153,537.3	32,174,344.9	484.0	32,328,366.2

The picture everyone has seen, now measured



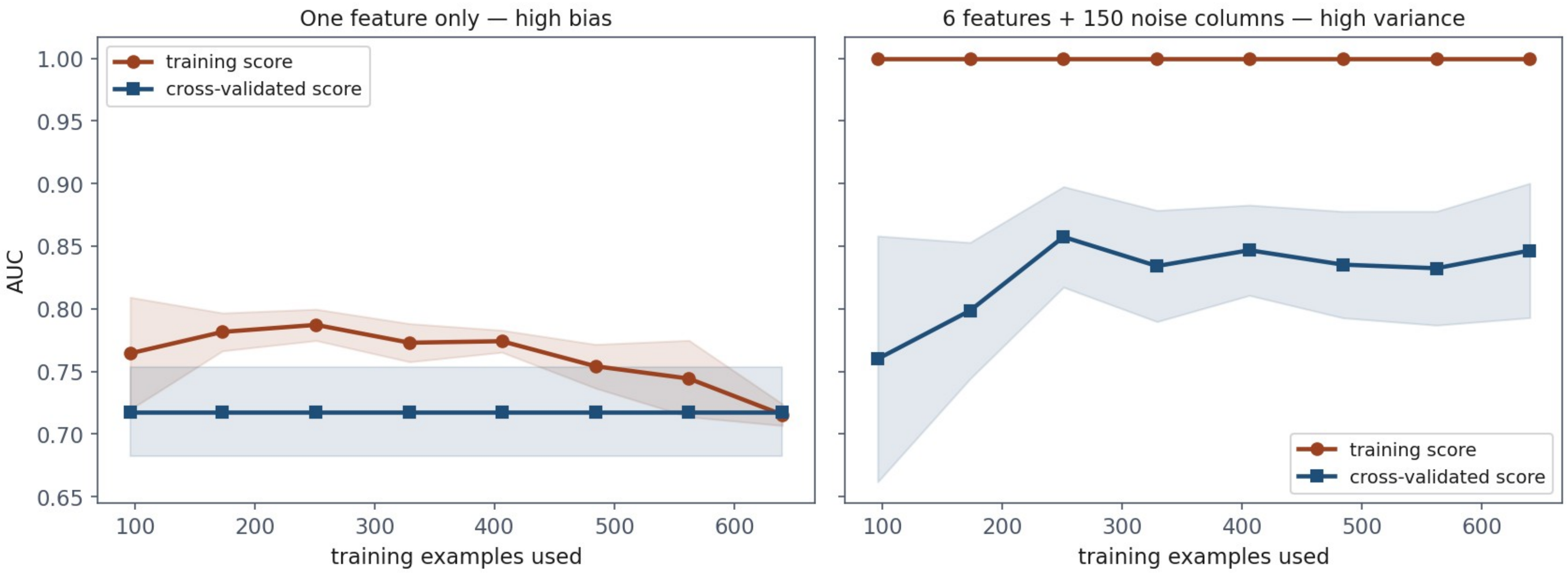
The floor you cannot go below

- Noise variance is **484** at every complexity
- A reported error below the floor means **contamination**, not excellence

The best model depends on how much data you have

Degree	m = 25	m = 60	m = 150
2	**614.3**	570.6	555.3
5	861.8	**540.9**	**506.9**
12	32,328,366.2	7,626.9	899.0

Learning curves: the same diagnosis, without knowing the truth



Which fix to buy

- Curves **meet low** → bias. More data will **not** help
- Wide gap, validation **still rising** → variance. More data **will**

Notebook 2, live

- Measuring bias and variance; the learning curves

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Now: what cross-validation does not protect you from

- 2,000 columns of **pure random noise**
- No relationship to the label whatsoever
- The honest score of any model on it is **0.500**

The code almost everyone writes

```
selector = SelectKBest(f_classif,
k=10).fit(noise, y)
chosen = noise.loc[:,
selector.get_support()]
cross_val_score(model, chosen, y,
cv=folds)
```

0.931

- Cross-validated AUC **0.931** on data with no signal
- Four of five folds between 0.93 and 0.98

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Fix it, and look what happens

- Selection moved **inside** the pipeline, refitted per fold
- 20 different fold seeds: **0.658**
- The truth is still 0.500

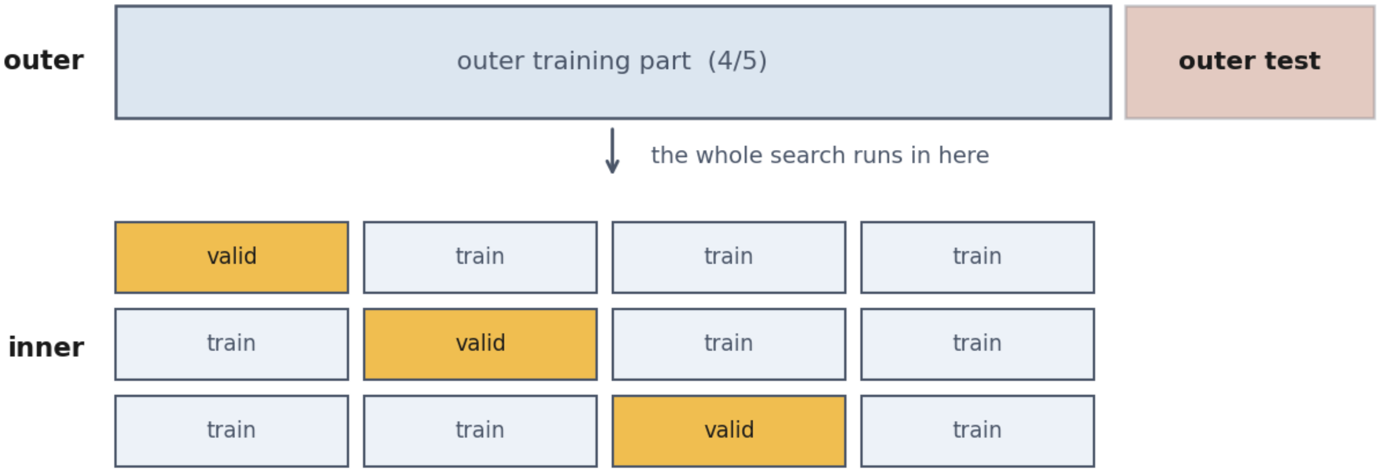
Why it does not return to 0.5

- With 2,000 columns and 800 rows, some columns match the label **by accident**
- That accident belongs to the **sample**, not to the split
- Every fold contains it

Hyperparameter search is the same problem

	AUC
best of 25 candidates	**0.7999**
average candidate	0.7265
nested cross-validation	0.6699
the truth	0.500

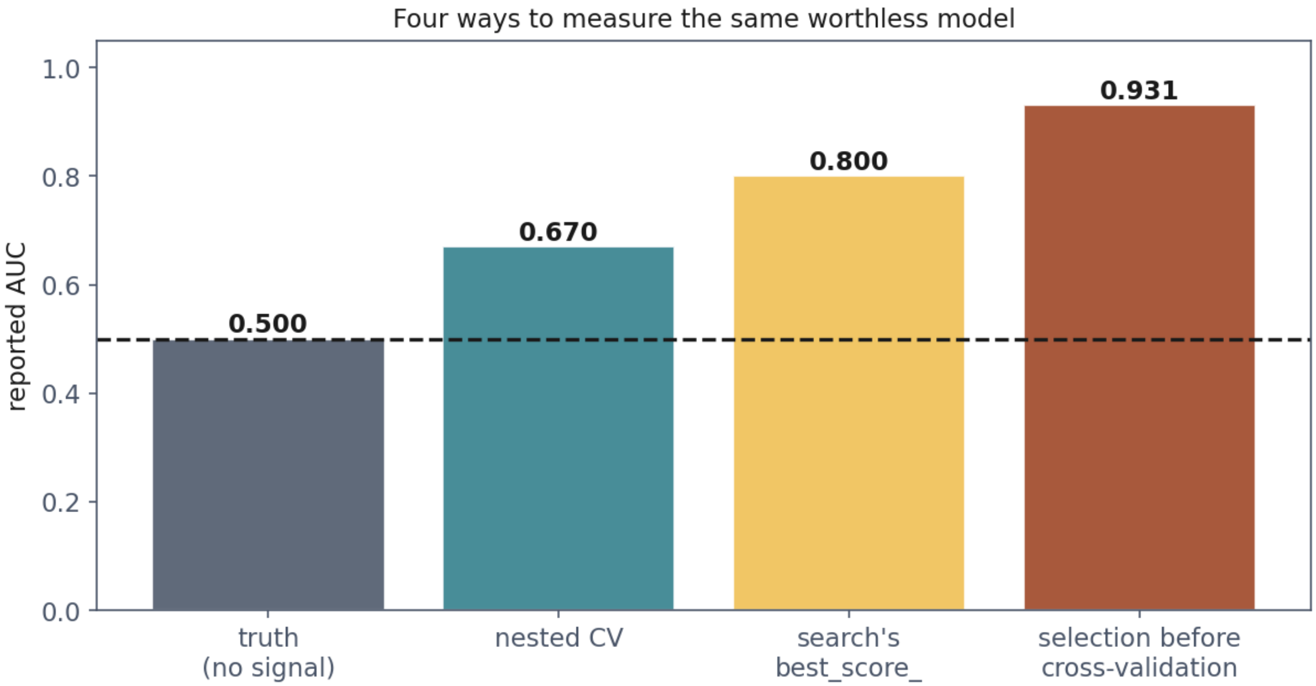
Nested cross-validation



The inner loop may overfit itself as much as it likes.

The outer test block was never part of it: that is the number you report.

Four ways to measure the same worthless model



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The debt, paid

Policy	Cost on the test set
threshold 0.50, the default	10,540 EUR
threshold chosen on **validation**	**4,700 EUR**
threshold chosen on **test**	3,300 EUR

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Three sets, three jobs

- **Train**: fit the parameters
- **Validation**: make every choice: model, hyperparameters, threshold
- **Test**: report. Touched **once**

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A fixed seed buys repeatability, not stability

- 30 seeds, all perfectly reproducible: **0.913 to 1.000**
- `AUC 1.000 (random_state=3)` is reproducible **and** misleading

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What to fix, and what to write down

- Every `random_state`, and `np.random.default_rng(seed)`
- **Library versions**: defaults change between releases
- What you could **not** fix: threads, GPU non-determinism

Notebook 3, live

- The 0.931, the fix, the nested search

What to put in a report

- The metric, the **spread**, and how it was estimated
- How many **positives** the test set held
- Every seed, and every library version
- Which choices were made on which data

What to take away

- **A test score is a measurement.** 0.885 to a perfect **1.000**, same model
- **Choosing on one split chooses noise**
- **bias² + variance + noise** is an identity, and the noise floor never moves
- **Curves that meet low mean bias**: more data will not help
- **Cross-validation protects against fold leakage, nothing more**

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Homework

- **Exercise 5**, discussed at the start of **Friday 30 October**
- A result that is too good. Find out why, then produce the honest number